MLOps Semester Project Specification & Dataset Report

Curriculum-Industry Skill Alignment Decision Framework for Identifying Employability Skill Gaps in CSE Graduates

Project Title:	Curriculum-Industry Skill Alignment Decision Framework	**Dataset Size:**	1,000 Graduates (20 Attributes)
Target Approach:	Regression Analysis (Skill_Gap_Index: 0.0 - 100.0)	**Domain:**	Computer Science & Engineering (CSE)
Generated Files:	`CSE_Graduate_Skill_Gap_Dataset.xlsx`	**Format:**	Styled Excel Workbook + PDF Documentation

1. Project Abstract & Framework Objectives

The rapid evolution of tech stack requirements in cloud computing, modern software engineering, and artificial intelligence has widened the gap between university academic curricula and industry hiring standards. This project implements an MLOps-driven decision framework that continuously evaluates graduating Computer Science & Engineering (CSE) students across academic, technical, soft skills, and practical experience dimensions.

By mapping these attributes into a continuous **Skill Gap Index (0.0 to 100.0)**, universities and recruitment teams can pinpoint structural weaknesses in curricula and deploy targeted training interventions before graduate placement cycles.

2. Dataset Schema & Attribute Specifications (20 Attributes)

The companion dataset contains 1,000 synthetic graduate profiles structured across 20 feature columns specifically engineered for regression modeling:

#	Attribute Name	Data Type	Range / Values	Description
1	**Student_ID**	Categorical	CSE_0001 - CSE_1000	Unique Primary Key
2	**CGPA**	Float	5.00 - 10.00	Cumulative Grade Point Average
3	**DSA_Score**	Integer	30 - 100	Data Structures & Algorithms Score
4	**System_Design_Score**	Integer	25 - 98	System Design & Architecture Evaluation
5	**DBMS_Score**	Integer	40 - 100	Database Management Systems Proficiency
6	**OOP_Proficiency**	Integer	35 - 100	Object-Oriented Programming Rating
7	**Cloud_DevOps_Score**	Integer	10 - 95	Hands-on Docker, Kubernetes & AWS/GCP
8	**Web_Dev_Score**	Integer	20 - 100	Full-Stack / Web Development Proficiency
9	**Problem_Solving_Rating**	Integer	800 - 2100	Competitive Coding Rating (LeetCode/ CodeChef)
10	**Aptitude_Score**	Integer	40 - 100	Quantitative & Logical Reasoning Assessment
11	**Communication_Rating**	Float	1.5 - 5.0	Soft Skills & Verbal Communication Score
12	**Teamwork_Score**	Integer	40 - 100	Behavioral & Group Collaboration Assessment
13	**Github_Projects_Count**	Integer	0 - 15	Verified Open-Source/Personal Repositories
14	**Internship_Duration_Months**	Integer	0, 2, 3, 6, 12	Total Industry Internship Experience

#	Attribute Name	Data Type	Range / Values	Description
15	**Certifications_Count**	Integer	0 - 4	Industry Certifications (AWS, GCP, CKA, etc.)
16	**Hackathons_Participated**	Integer	0 - 8	Hackathons / Coding Competitions Attended
17	**Mock_Interview_Score**	Integer	30 - 100	Technical & HR Mock Interview Performance
18	**Domain_Specialization**	Categorical	5 Categories	AI_ML, FullStack, DevOps, Cyber, DataEng
19	**Curriculum_Updated_Year**	Integer	2018 - 2024	Year University Curriculum was last revised
20	**Skill_Gap_Index**	Target (Float)	0.00 - 100.00	Regression Target (Higher = Larger Gap)

3. Target Generation Mathematical Formula

The continuous target metric ***Skill_Gap_Index*** represents the industry deficit. A lower score signifies high job readiness, whereas a higher score signifies an urgent requirement for remediation.

$$Skill\_Gap\_Index = 100 - (0.20 \times DSA) - (0.15 \times SystemDesign) - (0.15 \times MockInterview) - (0.10 \times CloudDevOps) - (2.5 \times Projects) - (2.0 \times Internships) - (4.0 \times Communication) - (2.0 \times CGPA) - 1.5 \times (CurriculumYear - 2018) + \varepsilon$$

4. Dataset Sample Preview (First 6 Student Profiles)

Student ID	CGPA	DSA	SysDes	Cloud	Projects	Intern	Comm	Domain	CurrYr	Gap Index
CSE_0001	7.11	76	65	73	1	3 m	4.2	AI_ML	2022	**10.54**
CSE_0002	9.59	41	37	47	3	12 m	2.4	DataEngineering	2019	**8.27**
CSE_0003	8.65	91	33	73	2	2 m	4.8	AI_ML	2024	**7.31**
CSE_0004	8.07	37	60	16	1	0 m	4.0	AI_ML	2022	**37.87**
CSE_0005	6.17	50	82	31	3	0 m	4.1	DevOps	2022	**23.86**
CSE_0006	6.17	99	78	90	5	2 m	2.3	DataEngineering	2018	**12.27**

5. MLOps Project Architecture & Workflow

1. **Data Ingestion & Tracking (DVC):** Store and track `CSE_Graduate_Skill_Gap_Dataset.xlsx` with Git & DVC.
2. **Model Experimentation (MLflow):** Train regression models (RandomForestRegressor, XGBoost) and log hyperparameters, MAE, RMSE, and R^2 scores.
3. **Explainability (SHAP Values):** Identify top features contributing to the skill gap score for curriculum revision decisions.
4. **Deployment & CI/CD:** Package the trained pipeline into FastAPI and deploy using Docker on Cloud (AWS/GCP).